Dynamic Programming

Part 3: Exercise 1 & 2 (week 15)

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Plan for today

- Time Iterations
- EGM
- Nonlinear grids

Blast-from-the-Pa(rt1)st: Deaton's model with finite horizon

Remember Deaton's model? with **Bellman equation**:

$$V_t(M_t) = \max_{C_t} \left\{ \frac{C_t^{1-\rho}}{1-\rho} + \beta \mathbb{E}[V_{t+1}(M_{t+1})] \right\}$$

s.t.

$$M_{t+1} = R(M_t - C_t) + Y_{t+1}$$

$$Y_{t+1} = \exp(\xi_{t+1})$$

$$\xi_{t+1} \sim \mathcal{N}(0, \sigma_\xi^2)$$

$$A_t = M_t - C_t \geq 0$$

With Euler Equation (necessary + sufficient here)

$$C_t^{-\rho} = \beta R \mathbb{E}_t[C_{t+1}^{-\rho}]$$

(derive with Bellman + Envelope Theorem, see week 9 slides)

Exercise 1-2: Intuition for Time Iterations (TI) and EGM

Both TI and EGM rely on backwards induction (in finite horizon)

$\rightarrow$ Nothing changed there!

In part 1, we used grid search for optimization:

For each state $\rightarrow$ Define grid of possible consumption, pick optimal

Problem: extremely slow, curse of dimensionality in choices

New thing: Rely on a bit more structure $\rightarrow$ Euler Equation:

TI: Call root-finder to solve $C_t^{-\rho} - \beta R \mathbb{E}_t[C_{t+1}^{-\rho}] = 0$

EGM: Introduce **post-decision state** $\rightarrow$ savings A_t

$\rightarrow$ Can compute RHS of Euler Equation

$\rightarrow$ Can then back out C_t^* of LHS

Exercise 1-2: TI and EGM Algorithms

1. Last period T: "eat what is left" $\Rightarrow C_T^*(M_T) = M_T$
2. Go to period $T-1$ and...

<u>TI: Loop over state space of M_{T-1}:</u>

Call root-finder: $C_{T-1}^{-\rho} - \beta R \mathbb{E}_t[C_T^{*-\rho}(R(M_{T-1} - C_{T-1}) + \exp(\xi))] = 0$

<u>EGM: Loop over post-decision state space A_{T-1}</u>

A_{T-1} known $\rightarrow$ Compute $M_T = RA_{T-1} + \exp(\xi^{gauss})$

M_T "known" $\rightarrow$ Compute $C_T^*(M_T)$

$C_T^*(M_T)$ "known" $\rightarrow$ Compute $\beta R \mathbb{E}[C_T^{*-\rho}(M_T)]$

Back out $C_{T-1} = \left(\beta R \mathbb{E}[C_T^{*-\rho}(M_T)]\right)^{-\frac{1}{\rho}}$

Infer M_{T-1} grid: $M_{T-1} = C_{T-1} + A_{T-1}$ (endo. grid)

For period $T-2, T-3, \ldots, 1$: Do the same as above

Exercise 1-2: TI and EGM, important detail

In first slide I presented Euler Equation:

$$C_t^{-\rho} = \beta R \mathbb{E}[C_{t+1}^{-\rho}]$$

It should actually be:

$$C_t^{-\rho} \geq \beta R \mathbb{E}[C_{t+1}^{-\rho}]$$

Why? Model has no-borrowing contraint: $A_t = M_t - C_t \geq 0$

Low $M_t \rightarrow$ Optimal to "move consumption" to today from future

$\rightarrow$ No-can-do! $\rightarrow C_t$ will be low $\rightarrow C_t^{-\rho} > \beta R \mathbb{E}[C_{t+1}^{-\rho}]$

With TI: If $C_t > M_t \rightarrow$ Set $C_t = M_t$

With EGM: Handle $A_t = 0 \Rightarrow C_t = M_t$ seperately.

$\rightarrow$ Linear policy $C_t^*(M_t) = M_t$ when $A_t = 0$

Grids, grids, linear grids, nonlinear grids

We do everything on grids, often: `np.linspace(`$M_{min}, M_{max}, \#M$`)`

Policy function is typically concave with:

$\rightarrow$ Lots of curvature when $M \rightarrow 0$

$\rightarrow$ Larger (linear) interpolation error for $M \rightarrow 0$

(Potential) fix: `nonlinspace` in `tools.py`

Make grid $\{y_i\}_{i=1}^n$ by discretizing $[x_{min}, x_{max}]$

Boundaries: $y_1 = x_{min}, y_n = x_{max}$

Inner: loop over $i = 1, 2, \ldots, n-1$ and fill in $y_i = y_{i-1} + \Delta_i$

$$\Delta_i = \frac{x_{max} - y_{i-1}}{(n-i)^\phi}, \phi \geq 1 \quad \phi \uparrow \rightarrow \text{More nonlinear grid}$$

E.g: $\Rightarrow \Delta_1 = \frac{x_{max} - x_{min}}{(n-i)^\phi}, \ \Delta_{n-1} = x_{max} - y_{n-2}$

Your time to shine!